

Project Specification: AI Thought Refinement Assistant

1. The Problem

Ideas often appear while studying, reading research papers, watching technical videos, or working on software projects. These thoughts are usually captured quickly in natural language, making them scattered, difficult to organize, and hard to build upon later.

This project aims to solve that problem by providing an AI assistant that transforms messy, unstructured thoughts into structured knowledge while preserving the original meaning. Instead of generating new knowledge, the AI focuses on refining and organizing the user's own thinking.

The generated knowledge can then be stored, connected, and expanded over time inside an Obsidian knowledge base.

2. Key Features

Conversational Idea Capture

The user sends a raw idea through Telegram using natural language.

AI Thought Refinement

The AI analyzes the user's message and transforms it into a clear, well-structured Markdown note. Rather than simply summarizing the text, it identifies the core idea, organizes supporting points, and preserves the user's original intent.

If information is incomplete, the AI explicitly lists missing details as **Open Questions** instead of generating assumptions or hallucinating missing content.

Automatic Markdown Generation

Each idea is converted into a consistent Markdown template containing:

- Title
- Summary
- Key Points
- Tags
- Open Questions (when applicable)

Obsidian Integration

The generated Markdown note is automatically saved into an Obsidian vault, allowing knowledge to grow incrementally instead of remaining as isolated notes.

Basic Knowledge Linking

When possible, the application creates simple connections between related notes using shared tags or keywords, without relying on embeddings or vector databases.

3. Constraints (MVP Scope)

To keep the project realistic for a one-week implementation, the first version will include the following limitations:

- Telegram text messages only (no voice support).
 - Single-turn interaction; the AI processes one message at a time without follow-up conversations.
 - Knowledge linking relies only on simple keyword or tag matching.
 - No cloud-hosted database or user authentication system (all notes are processed and stored locally).
 - The application focuses on organizing ideas and generating structured notes only.
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Expected Outcome

By the end of the week, the application will allow users to send raw ideas through Telegram and automatically convert them into clean, structured, and reusable Markdown notes stored inside an Obsidian vault. The result is a growing personal knowledge base that is easier to review, connect, and reuse for future learning, research, and software projects.